

Comprehensive Solutions Portfolio



Bulk Transcriptomics Solutions

RNA-seq (Eukaryotic Poly-A | With Reference Genome)

BT-EPAM-MDL

Applications

Method utilizes directional RNA libraries to process poly(A)⁺-enriched RNA into NGS libraries. Suitable for profiling changes in gene expression for biological discovery. The project can involve understanding differences between control vs treatment, or profiling changes across time-course or developmental series. Projects often use multiple biological replicates, analyzed to provide precision in inference of gene expression changes.

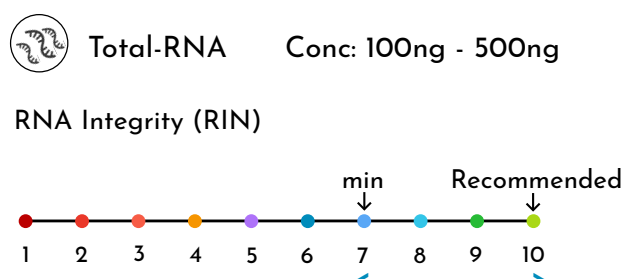
Pros

- High sensitivity for mRNA
- Accurate quantification of >98% of encoded mRNA
- Cost-effective
- Ideal for differential expression studies of mRNA to infer gene regulation

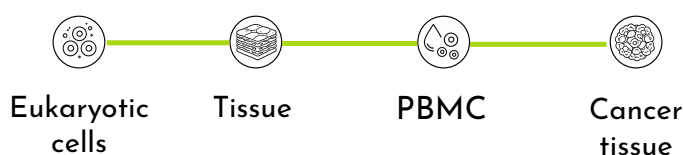
Notes

- Misses a small fraction of lncRNA, histone RNA, and non-polyadenylated RNAs
- Not suitable for degraded RNA due to 3' bias
- Reference genome/transcriptome required for read mapping

Input

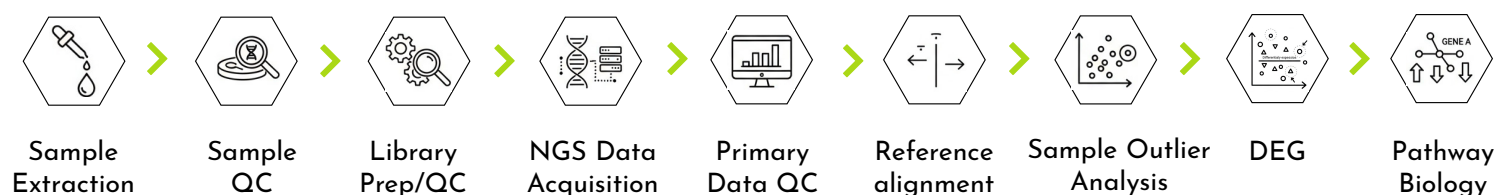


Sample Types



Platform Illumina platforms | 150 x 2

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Mapped data(.BAM) files sample and related QC reports
- Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
- Sample clustering reports
- Differential gene expression report
- Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
- Supporting visualization each step (PCA/UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathway maps*)

Add-ons

- *KEGG Pathview maps for pathways selected according to the interest of the study

RNA-seq (Eukaryotic Poly-A | Non-Model Organism - De Novo)

BT-EPAN-NMD

Applications

For efficient RNA seq Studies, Strain variations or lack of an available nearest reference require creation of a de novo transcriptome assembly. De novo transcriptome assembly improves read mapping and annotation, especially for plant and fungal samples where reference variation limits suitability. Suitable for projects where an isolate is characterized for changes across treatments or developmental stages.

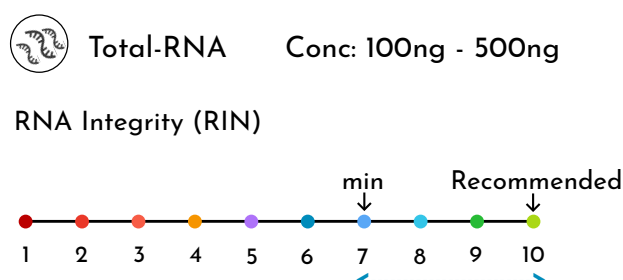
Pros

- Suitable for samples requiring reference genome creation
- High sensitivity for polyadenylated RNA
- Accurate expression quantification after assembly
- Suitable for evolutionary and ecological studies

Notes

- Higher computational complexity than reference-based RNA-seq
- Not suitable for highly degraded RNA

Input

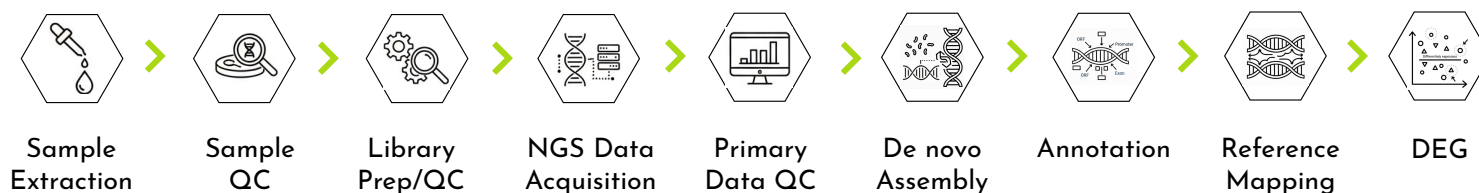


Sample Types



Platform Illumina platforms | 150 x 2

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- QC reports, mapping summary, alignment summary, and de novo statistics
- Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
- Sample clustering reports
- Differential gene expression report
- Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
- Supporting visualization each step (PCA/UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathway maps*)

Add-ons

- *KEGG Pathview maps for pathways selected according to the interest of the study
- Transcriptome secondary metabolite mapping

Applications

Suitable for comprehensive profiling of total RNA when requirements span non-polyadenylated RNAs such as histone RNA, tRNA, and other RNA species. The method uses rRNA depletion, followed by generation of directional RNA libraries for NGS profiling. Total RNA profiling is suitable for partially degraded RNA where RIN values do not meet criteria for poly(A)-based RNA profiling.

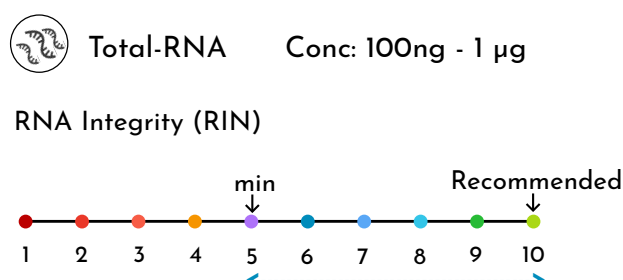
Pros

- Profiles total RNA, including tRNA, lncRNA, etc.
- Suitable for partially degraded RNA
- Provides broad transcriptome coverage
- Enables isoform-level quantification using a reference genome

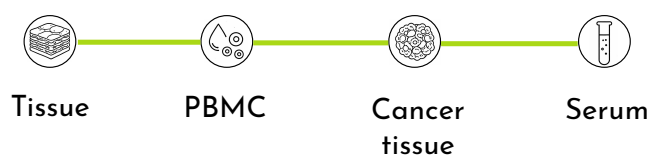
Notes

- Increased data complexity and higher sequencing depth required
- rRNA depletion efficiency impacts data quality

Input

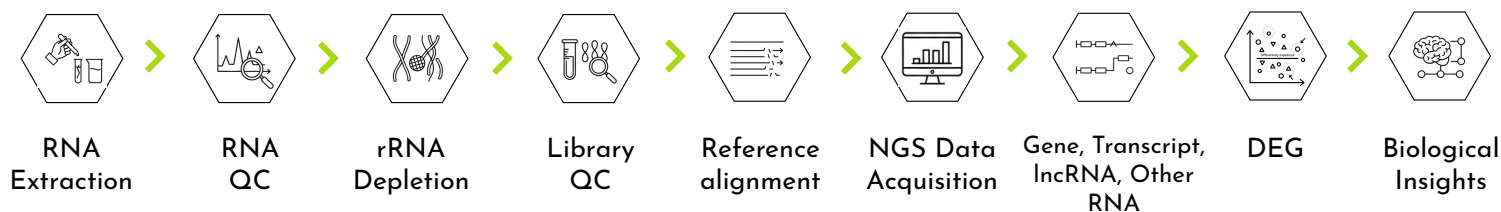


Sample Types



Platform Illumina platforms | 150 x 2

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Raw Data and Raw Data counts across all samples.
- Normalized read count for each samples along with Sample Grouping/Clustering and QC related plots : PCA plot, Heatmap based on top 2000 variant genes, dendrogram and Sample correlation plot.
- Differential expression reports for each comparison along with corresponding visualization - Volcano plots.
- Functional over-representation (enrichment) analysis per comparison, along with corresponding visualization - Gradient Bar plots for top 20 enriched GO: BP, MF, CC and Reactome Pathways.
- Raw Count and Normalized count of lncRNA for each samples along with Sample Grouping/Clustering and QC related plots : PCA plot, Heatmap on top 2000 variant genes, dendrogram and Sample correlation plot.
- Differential expression reports of known lncRNA for each comparison along with corresponding visualization - Volcano plots..
- Functional over - representation (enrichment) analysis for direct lncRNAs, cis -targets mRNA and mRNA interacting with lncRNA and their supporting plots and tables.

Add-ons

- Identification of co - expressed mRNA and lncRNA followed by visualization Interaction network for lncRNA and mRNA.
- Detection of Known and Novel circNAs with the supporting plots
- circRNA expression analysis per comparison. Volcano plot and Heatmap of Significant Circular RNAs
- Summary statistics of Circular RNA and their supporting plots
- Analysis - ASE (Alternative Splicing Event) per and Identification of Novel splice variants
- Overall Volcano plot for Differentially Expressed ASE and Individual Volcano plots for each events
- Scatter plots for showing splicing levels (percent - spliced -in, PSI) between two conditions.
- Gene ontology - Biological process, Molecular Function and Cellular components (overrepresentation analysis using enriched genes of top 10 differential alternative splicing events and their supporting plot.

RNA-seq (bacterial | With Reference Genome)

BT-BRNA-REF

Applications

Bacterial RNA-seq is used to profile gene expression changes upon treatment. RNA-seq enables sensitive detection of differential gene expression. Reference-based mapping can be variable and may require de novo assembly to discover transcripts, for example those acquired by horizontal gene transfer or encoded on plasmids/episomes. The process involves microbial rRNA depletion, followed by preparation of directional RNA libraries for NGS.

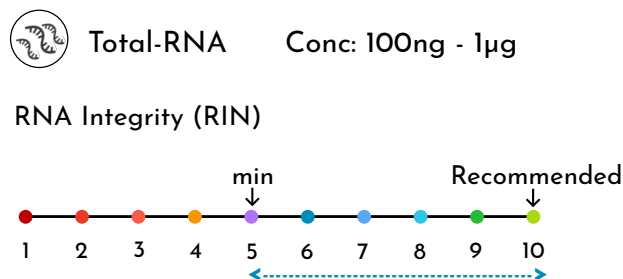
Pros

- Comprehensive coverage of coding bacterial RNAs
- High-accuracy reference-based alignment
- Suitable for operon- and pathway-level analysis
- Enables condition-specific expression profiling

Notes

- Requires high-quality RNA for effective rRNA depletion
- Reference genome quality impacts read mapping
- Not suitable for mixed/highly complex samples
- Reference-based mapping can miss novel transcripts, requiring de novo assembly.

Input



Sample Types



Platform Illumina platforms | 150 x 2

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Raw & QC Trimmed data (.fastq files) and QC reports per sample Mapped data (.BAM) files sample and related QC reports
- Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts Sample clustering reports Differential gene expression report
- Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
- Supporting visualization each step (PCA/UMAP/tSNE plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathview map)

RNA-seq (Blood | 3' End Tagging)

BT-BACT-BLD

Applications

Cost-effective, high-throughput RNA sequencing for gene expression profiling from blood samples using 3' end-tag RNA-seq, optimized for large-scale studies requiring robust expression quantification from whole blood. The method involves globin mRNA removal, followed by directional RNA library preparation. Blood RNA sequencing is suitable for whole-blood transcriptome profiling, particularly in biomarker discovery studies, where globin removal improves gene profiling and enhances coverage of low-abundance blood transcripts.

Pros

- Cost-effective for high sample numbers
- Optimized sequencing requirements to focus on genes of interest
- Robust gene-level quantification
- High reproducibility across samples

Notes

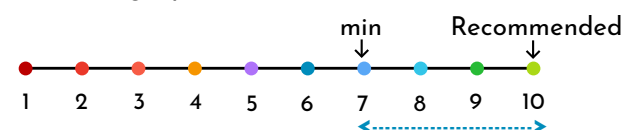
- Limited information on splice variants and isoforms
- Reference genome-dependent

Input



Total-RNA/Blood-derived RNA
Conc: 10ng - 100ng

RNA Integrity (RIN)



Sample Types



Whole Blood



Stabalized Blood
PAXgene/Tempus/Similar tubes

Platform Illumina platforms | 150 x 2

Process Map



Whole Blood
RNA Extraction



RNA QC



Globin
Depletion



Library
QC



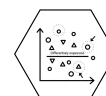
NGS Data
Acquisition



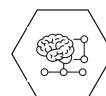
Primary
Data QC



Reference
mapping



DEG



Biological
insights

Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Raw & QC Trimmed data (.fastq files) and QC reports per sample
- Mapped data(BAM) files sample and related QC reports
- Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
- Sample clustering reports
- Differential gene expression report
- Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
- Supporting visualization each step (PCA/UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathway maps*)

Add-ons

- *KEGG Pathview maps for pathways selected according to the interest of the study

RNA-seq (FFPE | Human/ Mouse/ Rat | With Reference Genome)

BT-FFPE-HUM

Applications

Optimized RNA sequencing for formalin-fixed, paraffin-embedded (FFPE) human samples, enabling reliable gene expression profiling from degraded RNA. Designed for translational, clinical, and retrospective studies using well-annotated human reference genomes.

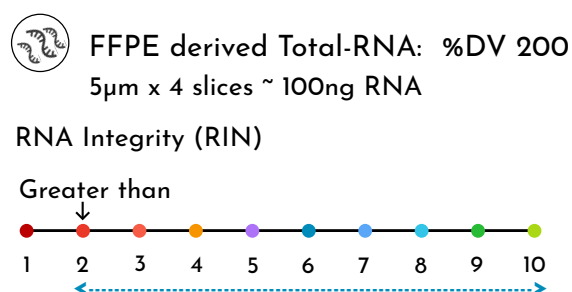
Pros

- Compatible with degraded and fragmented RNA
- Enables analysis of valuable archived samples
- Robust gene-level expression quantification
- Reference-based alignment ensures accuracy
- Ideal for clinical and translational research

Notes

- Limited isoform and splice-variant detection
- Lower complexity compared to fresh-frozen
- Quality dependent on fixation and storage conditions
- Reference-genome dependent

Input



Sample Types



Platform Illumina platforms | 150 x 2

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Raw data QC reports
- Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
- Sample clustering reports
- Differential gene expression report
- Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
- Supporting visualization each step (PCA/UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathway maps*)

Add-ons

- *KEGG Pathview maps for pathways selected according to the interest of the study
- Splice isoform / Fusion gene match / lncRNA

FFPE Cancer Transcriptome (Discovery RNA-seq | Human)

BT-FFPE-DIS

Applications

Discovery transcriptome is designed to provide multiple layers of information from a tumor sample. The method is optimized to work with fragmented RNA while maintaining broad transcript coverage. The discovery transcriptome is generated by multiplexing multiple libraries to address FFPE-associated biases. The computational methods enable detection of gene fusions, alternative splice forms, estimation of tumor signatures, tumor immune cell composition, and neoantigens.

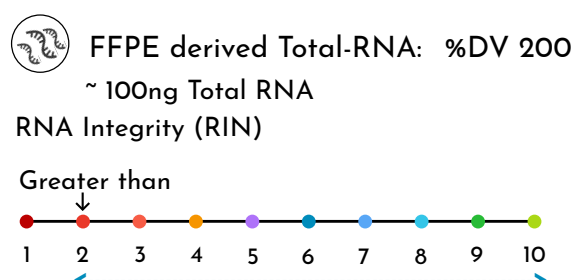
Pros

- Discovery research from archived FFPE samples
- Broad transcriptome coverage
- Optimized for fragmented RNA
- Reference-based analysis for high confident results
- Ideal for in detail tumor profiling

Notes

- RNA fragmentation may impact low-abundance transcripts
- FFPE sample quality leads to variations in discovery
- Reference-genome dependent

Input

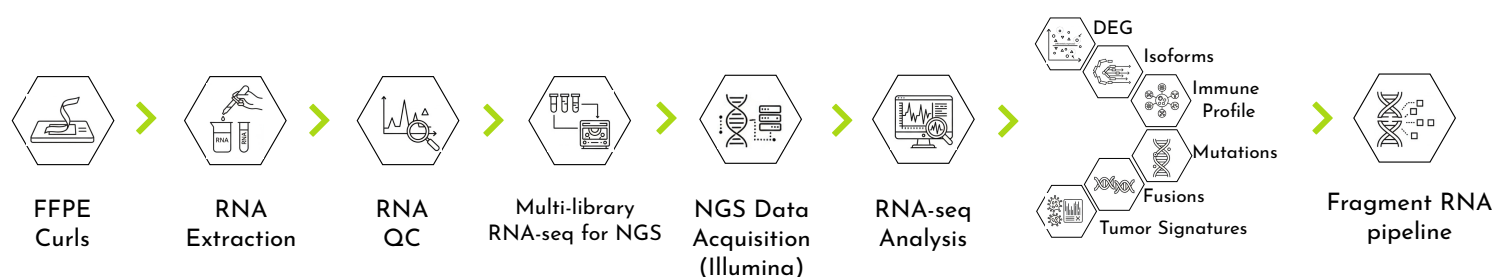


Sample Types



Platform Illumina platforms | 150 x 2

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
 - Data QC / Sequence Data Pre-processing Summary
 - Raw & QC Trimmed data (.fastq files) and QC reports per sample
 - Mapped data(.BAM) files sample and related QC reports
 - Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
 - Gene expression report, FPKM / TPM
 - Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
 - UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots
 - *KEGG Pathview maps for pathways selected according to the interest of the study
 - Splice forms, Transcripts, Fusion RNA
- Add-ons
- Tumor signatures, Immune profiling
 - Mutations from RNA-seq

RNA-seq (Plant / Microbial | Host-pathogen Co-expression)

BT-HPCE-PLT

Applications

The method utilizes multistage profiling of RNA, in eukaryotes with plant-fungal interaction to analyze the directional poly A enriched library for host and pathogen gene expression alteration, upon infection/treatment as single or multi-time point studies. When the pathogen is bacterial Total RNA seq with dual ribosomal RNA depletion is applied, prior to directional library preparation.

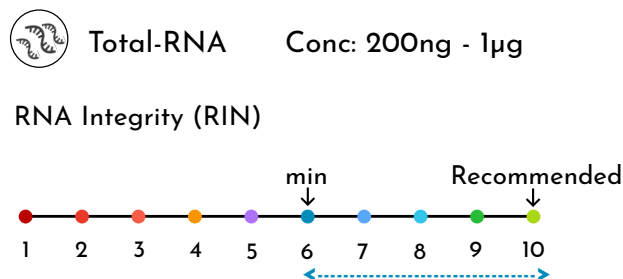
Pros

- Simultaneous host-pathogen transcript profiling
- Dynamic interaction and response signatures
- Cross species regulatory pathway discovery
- Efficient sample and sequencing utilization
- Ideal for discovery and hypothesis-driven studies

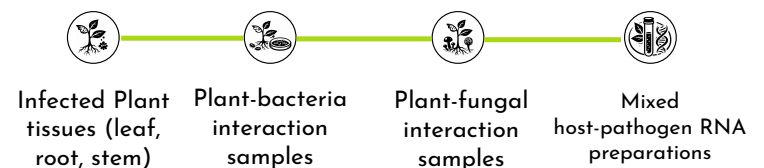
Notes

- Requires careful library design and sequencing depth
- Host RNA often dominates total reads
- Requires accurate reference genomes
- Complex data analysis and interpretation

Input

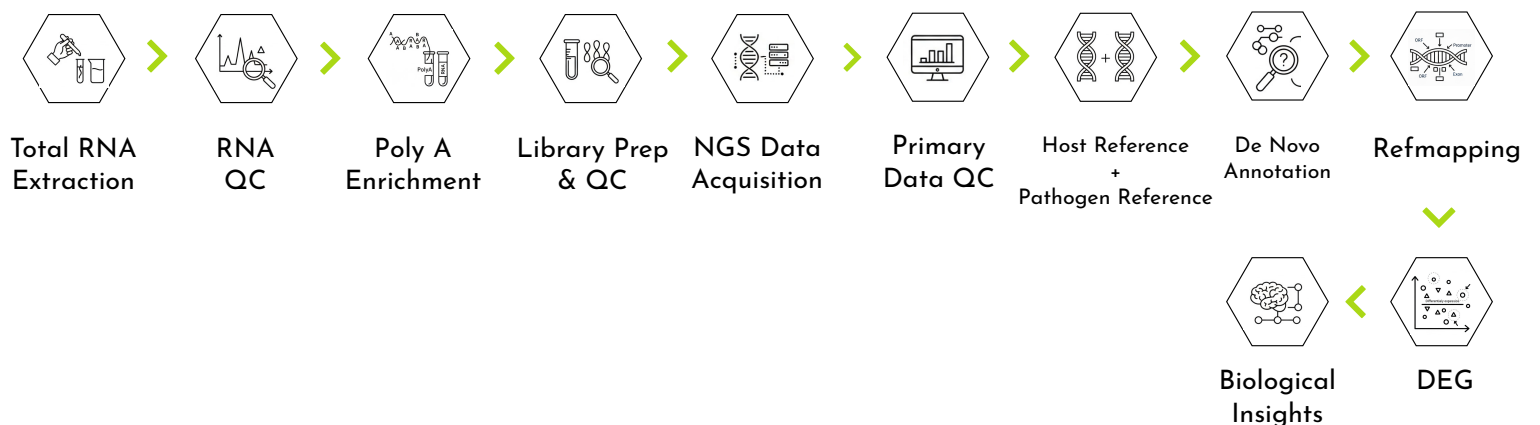


Sample Types



Platform Illumina platforms | 150 x 2

Process Map (Fungal-Host)



Process Map (Bacterial-Host)



Standard Deliverables

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Sequencing (Reference based*) - only protein coding genes - Raw & QC Trimmed data (.fastq files) and QC reports per sample
- Extraction of data for host and pathogen Mapped data (.BAM) files sample and related QC reports, host and pathogen
- Analysis performed separately for Host and pathogen : Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
Sample clustering reports
- Differential gene expression report
- Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways) Interaction between the pathogen and host transcriptome using WGCNA networks Protein - interaction network analysis
- Supporting visualization each step (PCA/UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathview maps* and many more..!)
- *KEGG Pathview maps for pathways selected according to the interest of the study *Reference genome/transcriptome must be known for the organism,

RNA-seq (Plant / Microbial | De Novo Transcriptome)

BT-DNVO-PLT

Applications

Plant and fungal isolates show significant variation across ecotypes and strains. To address this, we provide an option to generate a de novo transcriptome assembly, which is annotated and used for differential gene expression analysis. The process uses poly(A)+ RNA or total RNA, prepared as directional RNA libraries, to generate a de novo assembly. This approach supports both transcript discovery and gene expression profiling.

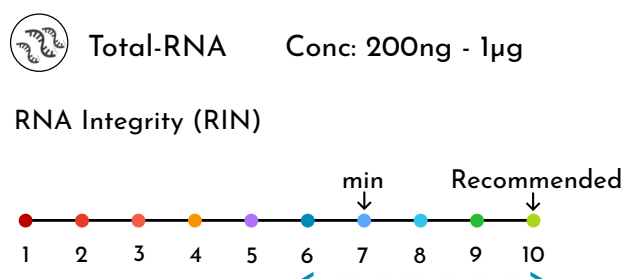
Pros

- No reference genome required
- Enables discovery of novel genes and transcripts
- Broad transcriptome coverage
- Suitable for emerging and understudied species
- Flexible for diverse plant and microbial systems

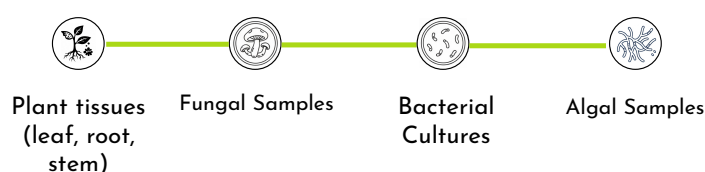
Notes

- Assembly quality depends on RNA quality and sequencing depth
- Higher computational complexity than reference-based RNA-seq
- Isoform resolution may vary across organisms

Input

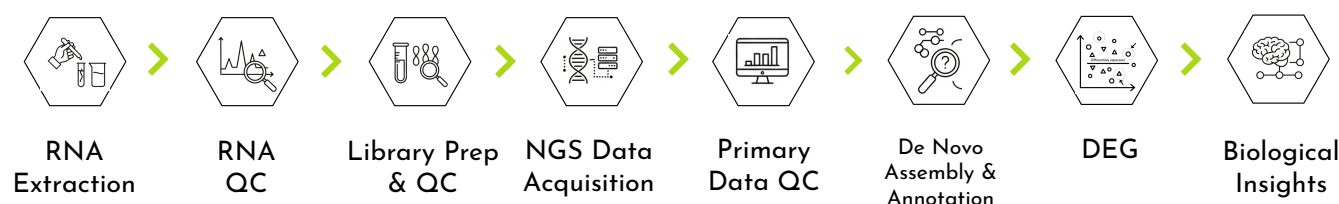


Sample Types



Platform Illumina platforms | 150 x 2

Process Map (Fungal-Host)



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Sequencing (Reference based*) - only protein coding genes - Raw & QC Trimmed data (.fastq files) and QC reports per sample
- Extraction of data for host and pathogen Mapped data (.BAM) files sample and related QC reports, host and pathogen
- Analysis performed separately for Host and pathogen : Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
- Sample clustering reports
- Differential gene expression report
- Functional enrichment reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTO ME pathways) Interaction between the pathogen and host transcriptome using WGCNA networks Protein - interaction network analysis
- Supporting visualization each step (PCA/UMAP plots, Heatmaps, Volcano plots, Functional enrichment gradient plots, KEGG pathway maps* and many more..!)
- *KEGG Pathview maps for pathways selected according to the interest of the study *Reference genome/transcriptome must be known for the organism,

LongRNAseq (Discovery)

BT-LRNA-DIS

Applications

Long RNA-seq uses cDNA converted from poly(A)-enriched RNA, sequenced using long-read sequencing. The use of long reads improves the ability to resolve structural variants (alternative isoforms, splice variants), long indels, and polyploidy-induced artifacts that are often missed in short-read assemblies. This method can be used to develop reference transcriptomes or for discovery studies.

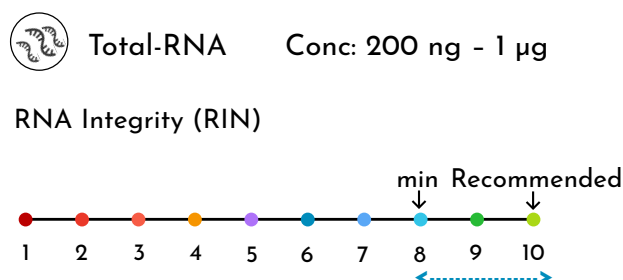
Pros

- Discovery of splice forms and isoform fusions
- Captures novel transcripts and isoforms beyond annotated genes
- Enables variant discovery and detection of alternative splicing and structure variation
- Suitable for polyploid genomes

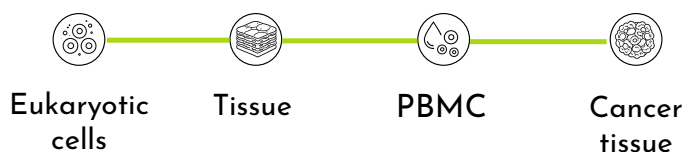
Notes

- Reads long cDNA; discovery depends on sequencing depth (million reads)
- Enriched via poly(A) capture
- Suitable for transcript discovery, not ideal for DEG analysis due to limited depth
- Does not detect RNA base modifications

Input

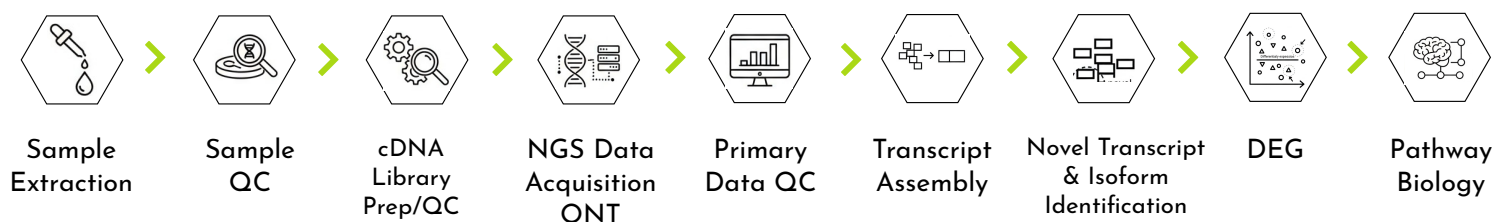


Sample Types



Platform ONT cDNA | 0.5 - 3 million⁺ Reads

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Sample QC, Library QC, Method Summary
- Mapped data(BAM) files sample and related QC reports
- Gene expression matrix (.csv/xlsx format) : Raw counts, Normalized counts
- Sample clustering reports: Transcript clustering, annotation, isoform, SNV/SNP
- Differential gene expression report
- Functional annotation reports (Gene ontology - Biological process, Cellular Component, Molecular Function & KEGG/REACTOME pathways)
- Supporting visualization: functional enrichment gradient plots, KEGG pathview maps*)
- *KEGG Pathview maps for pathways selected according to the interest of the study

miRNAseq NGS - Discovery (Non-Human)

BT-MIRD-NHU

Applications

Method utilizes small RNA-enriched library preparation to capture and sequence microRNAs (miRNAs) and other small regulatory RNAs from non-human samples. This discovery-driven approach enables identification, quantification, and comparative profiling of known and novel miRNAs across experimental conditions. miRNA discovery pipelines are applied to identify known and novel miRNAs, often from new isolates, strains, or treatment conditions. The workflow involves mapping to miRBase and computational models to annotate and discover novel small RNAs.

Pros

- Enables discovery of species-specific miRNAs
- High sensitivity for low-abundance small RNAs
- Provides insights into post-transcriptional gene regulation
- Ideal for comparative and evolutionary studies

Notes

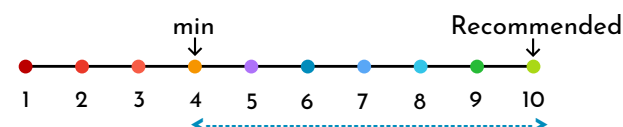
- Requires specialized small RNA library prep, totalRNA does not represent miRNA
- Functional interpretation relies on predicted miRNA targets Validation (qPCR/functional assays) recommended for novel miRNAs

Input

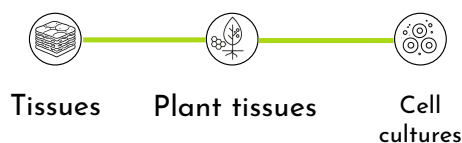


Total RNA / Small RNA-enriched RNA
Conc: 100 ng - 1 µg

RNA Integrity (RIN)



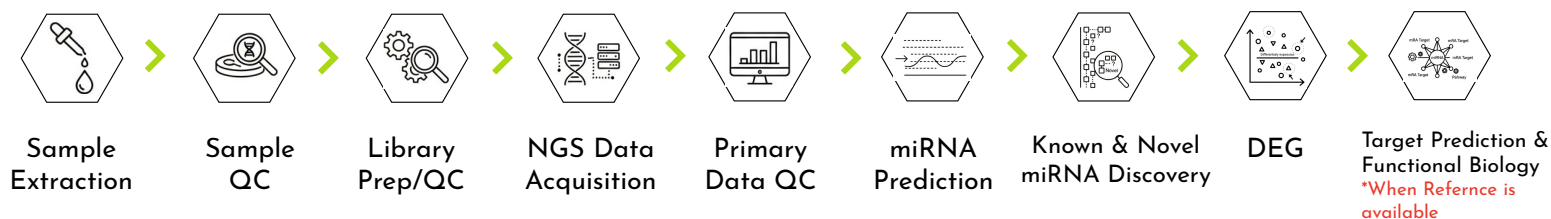
Sample Types



*RNA for small RNA, should not be prepared with total isolation-TROZOL/Mirvana

Platform Illumina platforms | 50 SE/75 SE/150 PE

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Raw Data QC and Raw Data counts across all samples.
- Normalized read count for each samples along with Sample Grouping/Clustering and QC related plots : PCA plot, Heatmap based on top 500 variant miRNAs. dendrogram and Sample - Sample correlation
- Known and Novel miRNAs identified by miRDeep2
- Differential expression reports for each comparison along with corresponding visualization - Volcano plots.
- Validated targets of Known miRNA identified by miRTarbase
- Functional over-representation (enrichment) analysis per comparison, along with corresponding visualization - Dot plots for top 20 enriched GO: BP, MF, CC and Reactome Pathways.
- Network for miRNA and mRNA (Validated targets)

Applications

Method utilizes small RNA-enriched, strand-specific library preparation to selectively capture microRNAs (miRNAs) and other small regulatory RNAs for next-generation sequencing. This workflow is optimized for comparative miRNA expression analysis across defined biological conditions. Multiple biological replicates are recommended to ensure accurate statistical inference of miRNA-level expression changes.

miRNA profiling is a method of choice to analyze total miRNA expression for discovery or validation. miRNA profiling enables identification of differentially modulated miRNAs across treatments, during development or disease progression, and in case-control cohorts. The method can be applied to infer functional modulation through target RNA analysis, and is ideally used paired with mRNA or total RNA sequencing.

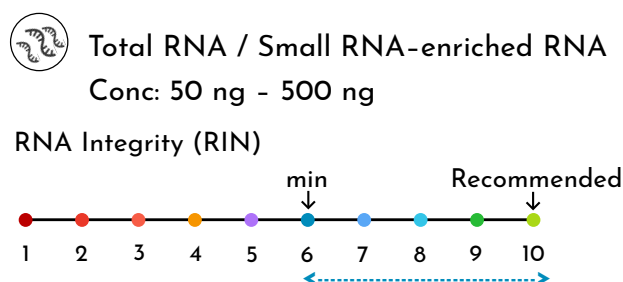
Pros

- Highly sensitive detection of miRNA expression changes on disease/treatments
- Accurate quantification of known miRNAs
- Ideal for biomarker discovery and validation studies

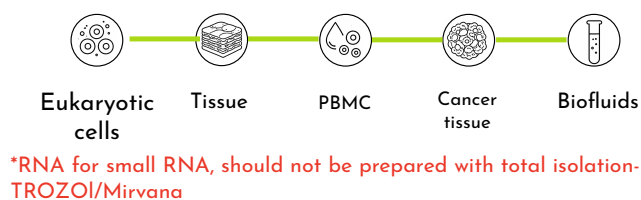
Notes

- miRNA profiling on treatment or disease
- Annotation on reference miRNA databases
- Functional insights are inferred via target prediction, ideal with paired mRNA/total RNA-seq

Input

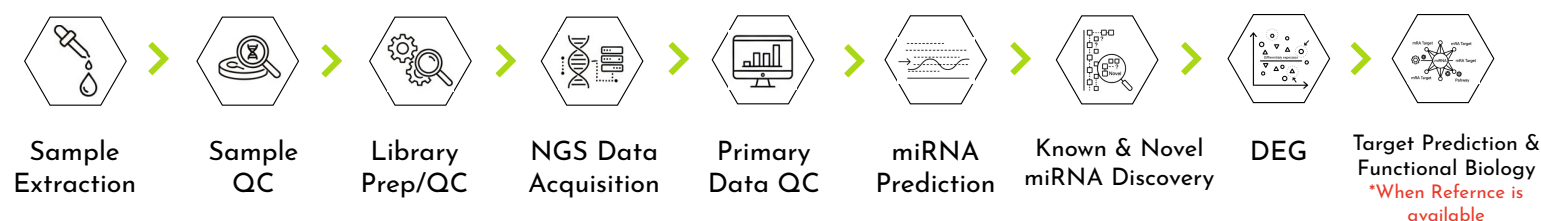


Sample Types



Platform Illumina platforms | 50 PE/50 SE/75 SE/150 PE

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
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- Normalized read count for each samples along with Sample Grouping/Clustering and QC related plots : PCA plot, Heatmap based on top 500 variant miRNAs. dendrogram and Sample - Sample correlation
- Known and Novel miRNAs identified by miRDeep2
- Differential expression reports for each comparison along with corresponding visualization - Volcano plots.
- Validated targets of Known miRNA identified by miRTarbase
- Functional over-representation (enrichment) analysis per comparison, along with corresponding visualization - Dot plots for top 20 enriched GO: BP, MF, CC and Reactome Pathways.
- Network for miRNA and mRNA (Validated targets)

miRNA-mRNA Integrated Analysis (Human / Mouse)

BT-MIRM-INT

Applications

Method integrates small RNA sequencing (miRNA-seq) and whole-transcriptome RNA sequencing (mRNA-seq) to comprehensively investigate post-transcriptional gene regulation. By jointly analyzing miRNA expression and corresponding mRNA targets, this approach enables identification of regulatory miRNA-mRNA interaction networks driving biological phenotypes. Multiple biological replicates are recommended to improve confidence in integrated regulatory inference.

The method uses total RNA extracted with compatible protocols such as TRIzol to generate parallel miRNA/small RNA libraries and total RNA or poly(A) RNA libraries. The co-expression and differential regulation of miRNAs and mRNAs in treated samples or case-control groups are analyzed as a network to unravel underlying biological mechanisms.

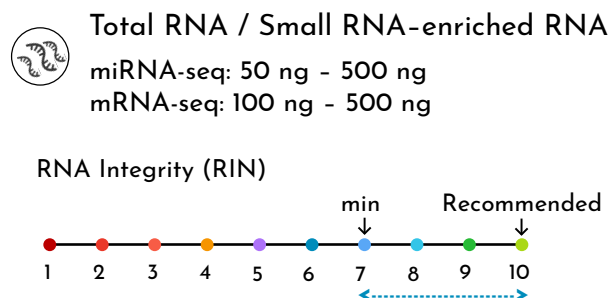
Pros

- Enables insight into miRNA gene regulation
- Integrates transcriptional and post-transcriptional layers
- Improves confidence in functional miRNA target identification

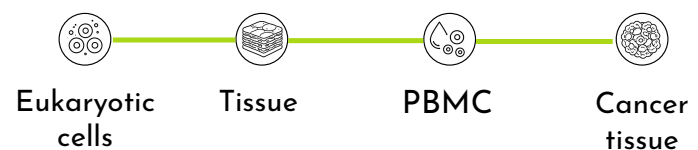
Notes

- Requires matched miRNA-seq and mRNA-seq samples
- Integrated analysis based on curated miRNA target databases
- Increased sequencing and analysis depth

Input

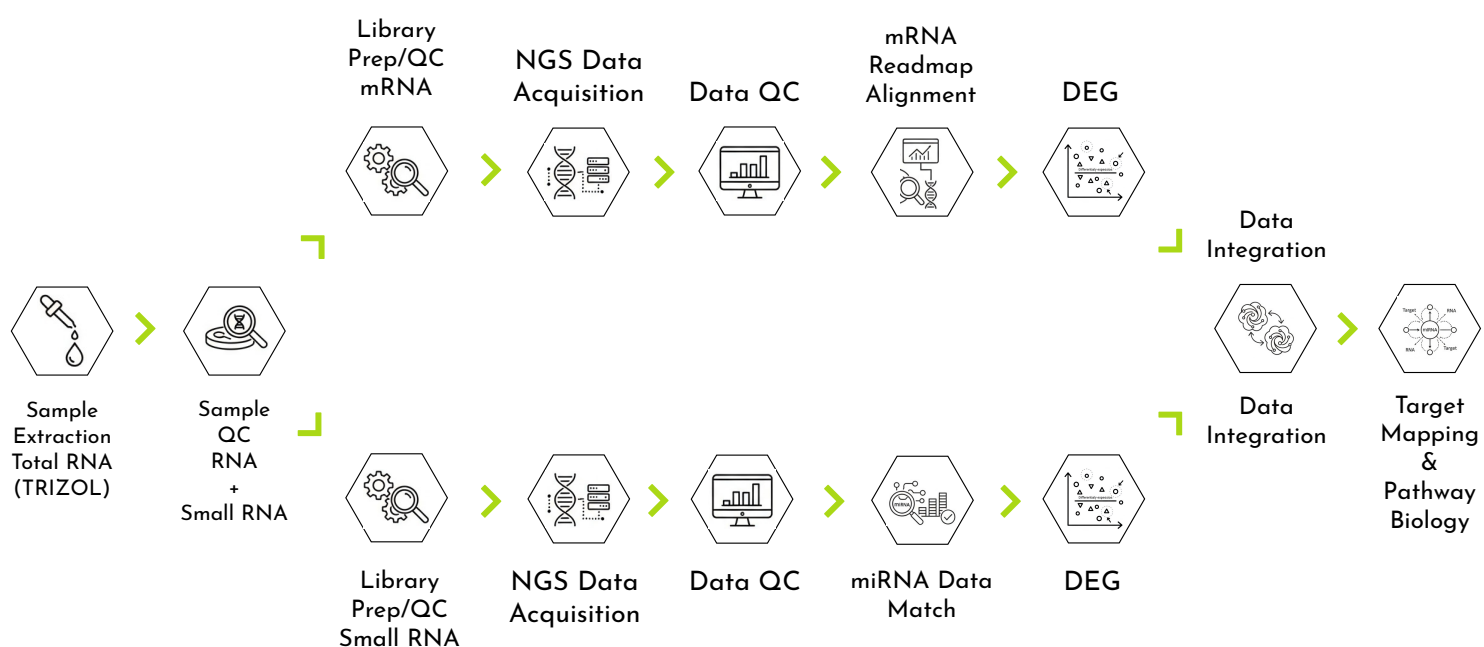


Sample Types



Platform Illumina platforms | 50 × 1 or 75 × 1
10M miRNA | 30-35M Total RNA

Process Map



Standard Deliverables

[Download Sample Report](#)

- Extraction QC, Sample QC, Summary of Methods
- Data QC / Sequence Data Pre-processing Summary
- Raw Data QC and Raw Data counts across all samples.
- Normalized read count for each samples along with Sample Grouping/Clustering and QC related plots : PCA plot, Heatmap based on top 500 variant miRNAs. dendrogram and Sample - Sample correlation
- Known and Novel miRNAs identified by miRDeep2
- Differential expression reports for each comparison along with corresponding visualization - Volcano plots.
- Validated targets of Known miRNA identified by miRTarbase
- Functional over - representation (enrichment) analysis per comparison, along with corresponding visualization - Dot plots for top 20 enriched GO: BP, MF, CC and Reactome Pathways.
- Network for miRNA and miRNA (Validated targets)



Metagenomics Solutions

Metagenome Human Gut V3-V4

MG-HGAS-VTF

Applications

Method utilizes targeted amplification and next-generation sequencing of the 16S rRNA gene V3-V4 hyper-variable regions to profile bacterial communities in human gut samples. This applied genomics approach enables accurate taxonomic characterization and comparative analysis of gut microbiome composition across biological conditions. The V3-V4 metagenome is a cost-effective and scalable method, where the primary objective is to measure changes in microbial diversity and analyze beta-diversity. These methods are ideal for characterized metagenomes with well-documented reference datasets, where the primary objective is understanding treatment differences within a cohort. The method is limited in resolving species- and genus-level taxonomy.

Pros

- High-resolution profiling of gut bacteria
- Cost-effective microbiome analysis
- Suitable for large cohort studies
- Robust taxonomic classification
- Ideal for comparative gut microbiome studies

Notes

- Limited to bacterial community profiling
- Species-level resolution may vary on match from V3-V4
- Functional insights are predictive
- Results influenced by sample handling and storage

Input



DNA ≥ 10 ng recommended



DNA Qubit > 10ng

Sample Types


Stool
samples


Swabs

*Stabilized samples | Fresh/Frozen not accepted

Platform Illumina platforms | 0.1 - 0.2M |
300 x 2

Process Map


Sample
Extraction

Sample
QC

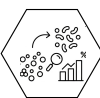
16S Amplicon QC
&
Library QC

NGS Data
Acquisition

Primary
Data QC

Data Clensing
ASV

QIIME
+ RDP

Taxonomic
Profile


β diversity

Standard Deliverables

[Download Sample Report](#)

- Sample & Library QC and summary
- Method & Workflow Summary
- Species-level diversity and abundance analysis using SILVA, Green-genes, and 16S-NCBI reference databases
- Read Clustering & Rarefaction Analysis
- Species/Genus Identification - Shannon, Simpson, and Chao diversity index
- PCA /PCoA plots
- Differential- Heatmaps
- KRONA Charts & Predicted Microbial Pathways (optional)

Metagenome Human Gut 16s Whole Amplicon

MG-HGAS-WHO

Applications

Method utilizes full-length 16S rRNA gene (full amplicon) sequencing to profile bacterial communities and improve taxonomic resolution. By capturing the complete 16S gene, the approach enables more accurate genus- and species-level classification compared to partial-fragment sequencing. Compared to the V3-V4 or other fragment-based methods, long amplicon 16S reads are matches bacterial species and genus. The data provide higher confidence in the discovery of species and inference of microbial pathways, leading to a superior estimate of alpha diversity. This method can be applied to well-referenced metagenomes such as the human gut, as well as environmental samples.

Pros

- Superior resolution 16S for diversity analysis
- Enhanced species-level identification
- Reduced primer-region bias
- Ideal for detailed microbiome profiling
- Suitable for cohort and longitudinal studies
- species resolution is desired

Notes

- Focused on bacterial community analysis
- Functional insights are predictive, not direct
- Species-level resolution depends on reference databases
- Sensitive to DNA quality and extraction/amplification

Input



Microbial DNA ≥ 10 ng recommended



DNA Qubit > 10ng

Sample Types



Stool samples



Swabs



Gut-derived clinical samples

Platform ONT sequencing platforms |
Long reads | 50K - 100K

Process Map



Sample Extraction



Sample QC



16S Amplicon QC & Library QC



NGS Data Acquisition



Primary Data QC



RDP/Silva Match



Species Abundance Typing



Diversity & Microbial Pathway

Standard Deliverables

- Sample & Library QC
- Method & Workflow Summary
- Raw Data QC & Processing
- Read Clustering, Matching, and Rarefaction Analysis
- Species/Genus Identification with Shannon/ Simpson Indices
- Differential Taxonomic Abundance with Heatmaps
- KRONA Charts & Predicted Microbial Pathways (optional)

[Download Sample Report](#)

Applications

Method utilizes integrated amplicon sequencing of bacterial and fungal marker genes to comprehensively profile mixed microbial communities within a single study. Bacterial communities are characterized using 16S rRNA gene sequencing, while fungal communities are profiled using ITS (ITS1/ITS2) or 18S rRNA regions. This integrated approach enables simultaneous assessment of bacterial-fungal composition, diversity, and co-occurrence patterns across biological or environmental conditions. The use of long amplicons provides the ability to type microbial as well as fungal/plant samples present in a microbiome. These investigations are well suited to profile species richness, diversity, and abundances in microbiomes to infer changes in time-course or treatment-dependent studies. Inference of alterations in microbial abundances provides clues to shifts in microbial pathways.

These methods are suitable when host DNA is high, sample DNA is limiting, or as a pre-pilot for whole-genome, metagenome, or metatranscriptome investigations.

Pros

- Simultaneous profiling of bacterial and fungal communities
- Enables cross-kingdom interaction analysis
- Improved ecological and functional interpretation
- Suitable for complex microbiome studies

Notes

- Based on targeted marker gene sequencing
- Functional insights are predictive
- Taxonomic resolution depends on reference databases
- Sensitive to sample quality and inhibitors
- ITS- may not detect all fungal types with equal efficiency

Input

 Microbial / Environmental DNA
≥ 10 ng recommended

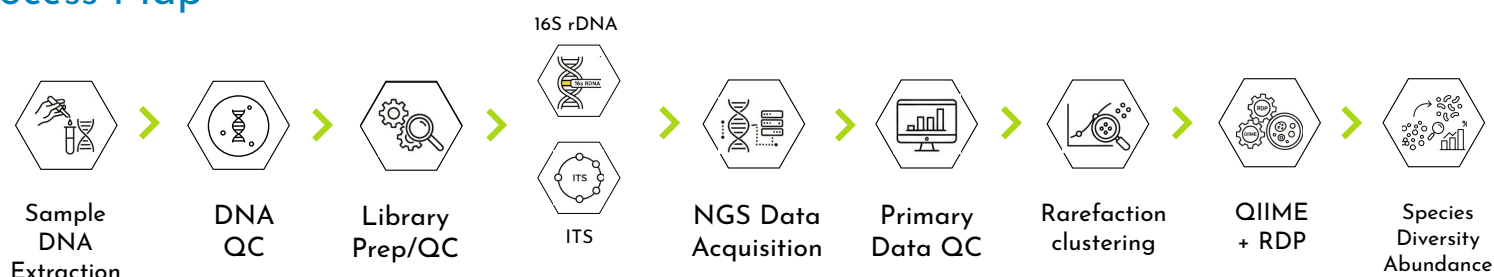
 DNA Qubit > 10ng

Sample Types



Platform ONT Long-read sequencing | 200K-250K reads 16s | 50K reads ITS

Process Map



Standard Deliverables

- Sample QC, Library QC, Summary of methods
- Data QC / Sequence data preprocessing summary
- 16S species match for typing and abundances
- ITS match for fungal and abundances
- Rarefaction Plots
- Species summary and counts
- Measure of diversity with Shannon & Simpson Indices
- When multiple samples in time covers / treatment are provided in Beta diversity, differential heatmaps and PCA plots provided

[Download Sample Report](#)

Whole Metagenome Sequencing - ONT

MG-WMGS-ONT

Applications

Method utilizes shotgun whole metagenome sequencing using Oxford Nanopore Technologies to comprehensively profile the genetic content of complex microbial communities. Long-read sequencing enables improved genome assembly, strain-level resolution, and detection of structural variants, mobile genetic elements, and antimicrobial resistance genes. This approach is suitable for bacteria, archaea, fungi, and viruses within a single experiment.

Whole-genome metagenomics offers the ability to discover novel microbial genomes and new genes/enzymes from complex microbiomes, where some key microbial communities may be un-culturable. The use of long-read data ensures reduced assembly bias and minimizes sequence contamination from closely related microbial genomes. The data are suitable for annotating microbial pathways and inferring mechanisms within microbial communities.

Samples extracted from host tissue may be mixed with significant amounts of host genomic DNA. High host background can skew sequencing results and may require host depletion, enrichment strategies, and/or additional sequencing depth to achieve the required rarefaction.

Pros

- Long reads enable improved genome assembly
- Enhanced strain- and species-level resolution
- Detects plasmids, phages, and structural variants
- Suitable for complex and mixed communities
- Enables comprehensive functional profiling

Notes

- Sensitive to sample prep and inhibitors
- Requires high-quality, high-molecular-weight DNA
- Bio-informatics analysis is computationally intensive
- Sequencing depth impacts genome recovery
- WG-metagenome has lower sensitivity compared to 16S for β -diversity
- infected microbiome analysis approach is recommended before metagenome approach

Input



High-molecular-weight Metagenomic DNA
≥ 1 µg recommended

Sample Types



Soil



Extreme
environment
samples



Biofilms and
sludge



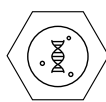
Water

Platform ONT Long-read sequencing platforms |
5 - 10 GB data on long reads

Process Map



Sample
DNA
Extraction



DNA
QC



Library
Prep/QC



NGS Data
Acquisition



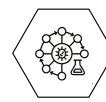
Primary
Data QC



Sequence
Assembly



Gene
Discovery
Annotation



Microbial
Pathway

Standard Deliverables

[Download Sample Report](#)

- Sample QC, library QC, and method summary
- Raw data QC, adapter trimming, and preprocessing
- Metagenome assembly summary
- Gene annotation and species/genus/strain-level matching against reference databases
- Identification of phages, episomes, and viral DNA with species rarefaction and abundance profiling
- Alpha diversity analysis, Shannon/Simpson indices)
- Beta diversity and inter-sample diversity estimation (for multi-sample studies)
- Host DNA considered as contamination and ideally removed during sample preparation

Whole Metatranscriptome Sequencing

MG-WMTS-ONT

Applications

Method utilizes shotgun RNA sequencing of total community RNA to capture the actively expressed genes of complex microbial communities. By sequencing and analyzing community-wide transcripts, this approach reveals functional activity, metabolic pathways, and regulatory responses in real time. The workflow typically includes rRNA depletion followed by cDNA library preparation and NGS. Metatranscriptomics profiles the microbial transcriptome community, providing key insights into the expressed functional genetic complement. The analysis of metatranscriptomes can be applied to discover encoded enzymes and novel pathways that are activated under specific conditions, such as bio-transformation and material degradation. It can also be used to infer metabolic switches when multiple time points are profiled to assess differential gene expression. The use of long-read methods for cDNA sequencing reduces assembly-induced artifacts arising from closely related species in mixed microbial communities.

Pros

- Captures active microbial functions
- Reveals real-time metabolic activity
- Complements metagenome sequencing
- Suitable for functional ecology studies
- Enables condition-specific pathway analysis

Notes

- RNA is highly sensitive to degradation from samples
- rRNA depletion from samples may be needed
- Sensitive to preparation and inhibitors
- For host + microbial systems, only expressed genomes are profiles unlike whole genome/ metagenome

Input



Total Community RNA

≥ 500 ng recommended



Nanodrop 230/260 |260/280

Sample Types



Soil



Extreme
environment
samples



Biofilms and
sludge

Platform ONT Long Read Platform |
10 - 15 GB data

Process Map



Standard Deliverables

- Sample QC, Library QC, Summary of Methods
- Data QC / Sequence Data pre-processing Summary
- Metatranscriptome Assembly
- Gene Identification, Species/Reference Abundance Estimates
- Identification of Transcripts-Pathway
- For Experiments with Time Course, Differential Expression to Analyze Modulation of Transcript and Species Abundances

[Download Sample Report](#)

Whole Genome Sequencing - Microbe (De novo)

AG-WGMI-DNV

Applications

Method utilizes shotgun whole-genome sequencing to generate a de novo assembly of microbial genomes without reliance on a reference sequence. This approach enables complete characterization of bacterial, archaeal, and fungal genomes, including novel species or strains. De novo assembly allows identification of coding and non-coding regions, genomic rearrangements, plasmids, and mobile genetic elements.

Pros

- Enables discovery of novel genomes and genes
- High-resolution strain characterization
- Suitable for poorly characterized organisms
- Supports functional and evolutionary studies
- Identifies Episomes, Virions etc
- Base modification in DNA

Notes

- Requires high-quality genomic DNA
- Assembly quality depends on sequencing depth
- Complex genomes may require long-read support
- 16S/18S purity testing suggested

Input



High Molecular Genomic DNA

- Short-read sequencing: $\geq 400\text{ng}$ - $1\mu\text{g}$
- Long-read sequencing (if applicable): $\geq 1\mu\text{g}$
- DIN/Gel analysis suggested
- 1 - 10ng low input prep will not allow DNA base modification calling

Sample Types



Bacterial
isolates



Archaeal
isolates



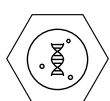
Fungal
isolates

Platform Illumina platforms / Long-read sequencing platforms |
100X Depth | 30X - 40X depth on long read

Process Map



Genomic
DNA
Extraction



Sample
QC



NGS
Library



NGS Data
(Longread)



Data QC



Sequence
Assembly



Gene
Discovery
Annotation



Genome
Annotation

Standard Deliverables

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, and pre-processing
- Genome Assembly Summary Statistics,
- Gene Calling, Gene Annotation
- Identification of Transposable Elements and Gene Clusters
- Antibiotic / Resistance Loci
- GFF3 File, Gene List, Annotation-Pathway Assignment, Gene Ontology Assignment
- CIRCOS Plots, Genome Maps
- *Where Applicable: Episome Assembly and Annotation

Additional option

- DNA methylation analysis

[Download Sample Report](#)

Whole Genome Sequencing - Fungal (De novo)

AG-WGFU-DNV

Applications

Method utilizes shotgun whole-genome sequencing to generate a de novo assembly of fungal genomes without reliance on a reference sequence. This enables comprehensive characterization of fungal genomes, including novel species or strains, complex gene families, repetitive regions, and secondary metabolite biosynthetic clusters. De novo assembly supports studies in fungal biology, pathogenicity, industrial strain development, and evolutionary genomics.

Pros

- Enables discovery of novel fungal genes
- Captures complex and repetitive genomic regions
- Suitable for poorly characterized fungal species
- Supports functional and evolutionary studies
- Structural variation, synteny and genome organization
- Repeat/Recombination events
- Haplotype phasing

Notes

- Requires high-quality genomic DNA of high molecular weight
- Assembly quality depends on genome size and complexity
- Long-read sequencing recommended for repeat-rich genomes
- 18S strain purity testing recommended

Input



Genomic DNA

- Short-read sequencing: ≥ 200 ng
- Long-read sequencing (recommended): ≥ 1 μ g
- DIN/Gel analysis for high molecular weight DNA

Sample Types



Fungal
isolates



Clinical fungal
isolates



Environmental
fungal isolates



Industrial and
agricultural fungal
strains

Platform Long-read sequencing platforms | 50-70X Long read |
~ 2-10 GB | Based on genome/ploidy estimates

Process Map



Genomic
DNA
Extraction



Sample
QC



NGS
Library



NGS Data
(Longread)



Data QC



Genome
Assembly



Gene
Discovery
Annotation



Genome
Analysis

Standard Deliverables

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, and pre-processing
- Genome Assembly Summary, Gene Detection, Genome Annotation
- SNP/Allele identification
- Haplotype information from long read parsing
- GFF3 File, Gene List, Repeat Elements, SINE/LINES, Pathway assignment, Secondary metabolite coding genes
- When multiple Strain/Strain matching scoped - synteny and comparative genome mapping for specific project types

[Download Sample Report](#)

References

- Estaki M et al. Curr Protoc Bioinformatics. 2020; PMID:32343490.
- Licata AG et al. Microbiol Spectr. 2025; doi:10.1128/spectrum.01673-25.

Whole Genome Resequencing (Eukaryotic)

AG-WGRS-EUK

Applications

Method utilizes high-coverage whole genome sequencing of eukaryotic organisms followed by read alignment to a reference genome to identify genomic variations. This resequencing approach enables accurate detection of single nucleotide variants (SNVs), insertions/deletions (InDels), copy number variations (CNVs), and structural variants across the genome when longread platform is used. Resequencing methods allow identification of individual mutants/variants and provides a whole genome unbiased discovery approach. The method can be used for generation of data for genetic analysis, discovery of novel rare SNP's, population structure analysis, discovery of coding and non-coding variations

Pros

- High-resolution genome-wide variant detection
- Accurate comparison against a reference genome
- Suitable for population and evolutionary studies
- Scalable from small to large genomes
- Supports trait and disease association analysis

Notes

- Requires a high-quality reference genome
- Structural variant detection benefits from higher coverage is recommended on long read
- Polyploid genomes require higher depth of coverage

Input



High Molecular Genomic DNA

≥ 1µg recommended - Long read
≥ 100ng recommended - Short read

- 1 - 100ng: Low input preparation for long read sequencing

Sample Types



Blood



Tissue



Eukaryotic
cells



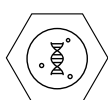
Plant/Fungi

Platform Illumina | ONT Long Read Platforms
30X - 40X based on genome Illumina;
Long Reads

Process Map



Genomic
DNA
Extraction



Sample
QC



NGS
Library



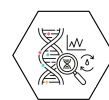
NGS Data



Data QC



Reference
Alignment



Variant
Calling



Variant
Annotation

Standard Deliverables

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, pre-processing, and Adapter trimming
- Reference Alignment Summary
- Variant Summary and Variant annotation
- Coding/Non-coding Variants
- Data pre-processing of VCF for genetic analysis based on project requirements

Additional options

- Comparison to individuals/ related reference genomes
- Extraction of variants for development of molecular marker

[Download Sample Report](#)

Whole Genome Resequencing (Microbial) AG-WGRS-MIC

Applications

Method utilizes high-depth whole genome sequencing of microbial isolates followed by alignment to a reference genome to identify genomic variations at single-nucleotide and structural levels. This resequencing approach enables precise detection of SNPs, InDels, copy number variations, and structural rearrangements, when long-read methods are applied.

Pros

- High-resolution strain-level variant detection
- Ideal for comparative and outbreak studies
- Accurate mutation analysis

Notes

- Requires a high-quality reference genome
- Closely related reference improves accuracy
- Sequencing depth affects low-frequency variant detection
- Structural variants benefit from long-read
- 16S/18S Strain purity testing recommended

Input



High Molecular Weight Genomic DNA

≥ 50 ng recommended- short read
>1µg- long read

- 10 - 100ng: low input preparation for long read sequencing

Sample Types



Bacterial
isolates



Archaeal
isolates



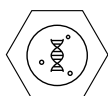
Environmental
microbial isolates

Platform Illumina | ONT Long Read Platforms
30X ~ 0.2 - 0.3 GB | 150 X 2
18X - 20X - Long Reads

Process Map



Genomic
DNA
Extraction



Sample
QC



NGS
Library



NGS Data



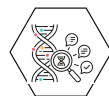
Data QC



Reference
Alignment



Variant
Calling



Variant
Annotation

Standard Deliverables

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, pre-processing, and Adapter trimming
- Reference Alignment Summary
- Variant Summary and Variant annotation
- Coding/Non-coding Variants
- Data pre-processing of VCF for genetic analysis based on project requirements

Additional options

- Analysis to specific datasets for pathway of mutated/modified genes
- Reference genomes assembly of unmapped reads

[Download Sample Report](#)

Applications

Method utilizes genome-wide DNA methylation profiling to identify and quantify epigenetic modifications associated with gene regulation and phenotype. The use of long read sequencing allows the analysis of modified bases. The method offer the ability to map individual sequences for multiple base modifications. DNA epimarks can be compared in different tissues/case-control in disease and in normal vs tumor DNA. Methylation analysis can be integrated with gene expression data for matched samples.

Pros

- Genome-wide discovery of methylation patterns
- High-resolution epigenetic profiling
- Suitable for disease and exposure studies
- Supports integrative multi-omics analysis

Notes

- Method requires non-modified genomic DNA
- PCR amplified DNA modifies and removes methylation
- Reference genome provides template to map methylated/modified DNA

Input



Genomic DNA

>500ng - 1µg recommended

High molecular weight, unmodified DNA recommended for genomic DNA quality analysis

Sample Types



Blood/
PBMC



Tissue



Cell lines



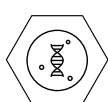
Plant/Fungi

Platform ONT Long Read platforms |
30 - 50X | Additional depth for polyploid genomes

Process Map



Genomic
DNA
Extraction



Sample
QC



NGS
Library



NGS Data



Data QC



Methylation
DNA
Basecalling



Reference
Alignment



Methylation
Annotation

Standard Deliverables

[Download Sample Report](#)

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, Pre-processing, and Adapter trimmimg
- Reference Alignment Summary
- Variant Summary and Variant annotation
- Coding/Non-coding Variants
- Data pre-processing of VCF for genetic analysis based on project requirements

Whole Genome De novo Sequencing - Large Genomes (> 500 Mb)

AG-WGDN-LRG

Applications

Method utilizes high-depth whole genome sequencing combined with advanced de novo assembly strategies to reconstruct large and complex eukaryotic genomes (> 500 Mb) without reliance on a reference sequence. This approach integrates short-read and long-read sequencing technologies to resolve repetitive regions, structural variations, and complex genome architectures. It is suitable for plants, animals, and other large-genome organisms, including non-model species.

Pros

- Enables assembly of large and complex genomes
- No reference genome required
- For repetitive and structurally complex regions
- Supports advanced functional genomics

Notes

- Requires high-quality, high-molecular-weight DNA
- Long-read sequencing strongly recommended
- High sequencing depth and coverage required for genome assembly

Input



High-molecular-weight Genomic DNA

- Long-read sequencing: $\geq 5\text{-}10\ \mu\text{g}$ recommended
- Short-read sequencing (polishing): $\geq 500\ \text{ng}$

Sample Types



Plant
Tissue

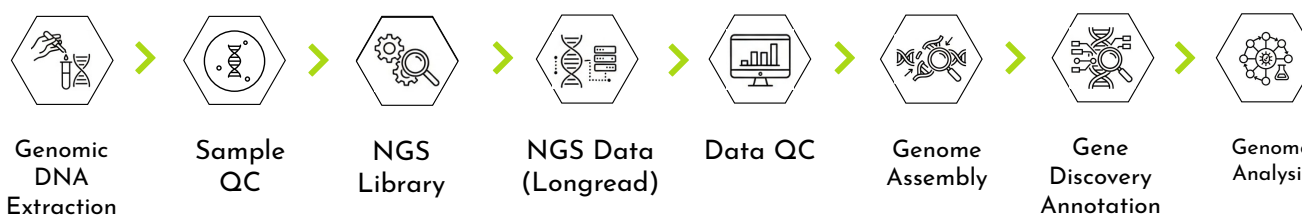


Animal
Tissue

Platform

Long-read sequencing platforms + Illumina platforms |
50 - 70X of estimated genome for 2n |
Additional depth for polyploid genomes

Process Map



Standard Deliverables

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, and pre-processing
- Genome Assembly Summary, Gene Detection, Genome Annotation
- SNP/Allele identification
- Haplotype information from long read parsing
- GFF3 File, Gene List, Repeat Elements, SINE/LINES, Pathway assignment, Secondary metabolite coding genes
- When multiple Strain/Strain matching scoped - Synteny and comparative genome mapping

[Download Sample Report](#)



Human Genetic Analysis

Applications

Method utilizes high-coverage whole genome sequencing (WGS) of genomic DNA to identify genetic variants. Sequencing reads are aligned to a reference genome followed by comprehensive variant calling and annotation. This approach enables detection of single nucleotide variants (SNVs), insertions/deletions (InDels), copy number variations (CNVs), and structural variants relevant to hereditary traits and disease predisposition. Whole genome variant analysis is the most cost-effective and lowest bias method for comprehensive analysis of an individual genome. The method is of choice for discovery of 'genetic disorders' in characterization of rare inherited disease. The data spans coding and non-coding intergenic regions and is suitable for analysis of copy number variations. A purpose built data annotation system is applied to interrogate the variants and provide up to date annotations, to rank variants by disease phenotypes, accelerating genetic analysis.

Pros

- Comprehensive genome-wide variant detection
- Identifies inherited disease-associated variants
- Suitable for clinical and research studies
- Supports family-based and population genetics
- High sensitivity and accuracy
- Variant annotation with curated datasets

Notes

- Requires high-quality genomic DNA
- Reference map to human reference
- Incidental findings may occur
- Clinical validation recommended for novel variants
- Repeat/long balanced variants may not be called with this application

Input



Genomic DNA (Germline)

DNA Quantity: ≥ 200 ng - 1 μ g recommended

High molecular weight DNA analyzed by gel/TapeStation

Sample Types



Blood



Tissue



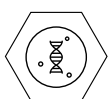
Buccal Swabs

Platform Illumina platforms | 30X - 35X | 150 x PE | 100 - 110 GB

Process Map



Sample DNA Extraction



DNA QC



PCR free Library Prep



NGS Data Acquisition



Data QC



Reference Alignment hg19, Grch38



Variant calling SNP, SNV-indel



CNV analysis



Variant annotation

Standard Deliverables

[Download Sample Report](#)

- Sample and Library Quality Control (QC)
- Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Identification and Summary Statistics
- Zygosity and Coding vs. Non-coding Variant Distribution
- Variant Annotation and Rare Variant Analysis (phenotype-driven, if provided)
- Pathogenic Variant Summary (if identified)
- Copy Number Variation (CNV) Analysis and Variant Effect Prediction

Exome Sequencing - Standard (Germline)

HG-EXOM-STD

Applications

Method utilizes targeted capture of protein-coding regions (exons) followed by high-throughput next-generation sequencing to identify germline genetic variants within clinically and biologically relevant genes. Sequencing reads are aligned to a reference genome, enabling accurate detection of single nucleotide variants (SNVs) and small insertions/deletions (InDels). Exome sequencing provides a cost-effective alternative to whole genome sequencing for studying inherited diseases, population genetics, and gene-phenotype associations.

Pros

- Protein-coding region
- High coverage of genes encoding proteins
- Well-established clinical interpretation frameworks
- Suitable for inherited disease studies as first line investigation of genetic disorders/population analysis

Notes

- Non-coding regions are not captured
- Structural variant detection is limited to short coding genes
- Coverage may vary across GC-rich regions
- Validation recommended for clinically relevant variants

Input



Genomic DNA (Germline)

DNA Quantity: ≥ 100 ng - 1 μ g recommended

Sample Types



Blood



Tissue



Buccal Swabs



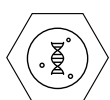
Cells

Platform Illumina platforms | 150 x 2 |
100 X Exome, 150 X 2 ~34.9 mb | 8 GB data

Process Map



Sample DNA Extraction



DNA QC



Capture + Library Prep



NGS Data Acquisition



Data QC



Reference Alignment



Variant annotation



Gene Rank by Phenotype



Reporting

Standard Deliverables

[Download Sample Report](#)

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Identification and Summary Statistics
- Zygosity and Coding, Variant Annotation and Variant Effect Prediction
- Pathogenic variants(if detected)
- Rare variants by specific phenotypes if described

Extended Exome Sequencing - Germline

HG-EXOM-EXT

Applications

Method utilizes enhanced target capture of protein-coding regions along with selected regulatory and clinically relevant non-coding regions to comprehensively profile germline genetic variants. Compared to standard exome sequencing, extended exome expands coverage to include untranslated regions (UTRs), splice sites, and disease-associated regulatory elements, improving variant discovery beyond canonical exons. Sequencing reads are aligned to a reference genome followed by variant calling and annotation.

Pros

- Broader coverage target than standard exome
- Improved detection of regulatory variants
- Higher diagnostic yield for rare diseases
- Suitable for unresolved clinical genetic research

Notes

- Intergenic regions out of capture not sequenced
- Structural variant detection is limited
- Coverage variability may occur in GC-rich regions

Input



Genomic DNA (Germline)

DNA Quantity: ≥ 100 ng - 1 μ g recommended

Sample Types



Blood



Tissue



Buccal Swabs

Platform Illumina platforms | 150 x 2
63 mb ~ 14 GB at 100 X

Process Map



Sample DNA Extraction



DNA QC



Capture + Library Prep



NGS Data Acquisition



Data QC & Preprocessing



Reference Alignment



Variant annotation



Gene Rank by Phenotype



Reporting

Standard Deliverables

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Identification and Summary Statistics
- Zygosity and Coding, Variant Annotation and Variant Effect Prediction
- Pathogenic variants(if detected)
- Rare variants by specific phenotypes if described
- *Note: For trio 37 MB | 33 GB data/ 11GB/Sample

[Download Sample Report](#)

Trio Genetic Analysis - Exome (Germline) HG-TRIO-EXO

Applications

Method utilizes exome sequencing of a parent-parent-child trio to identify inherited and de novo germline variants underlying genetic disorders. Protein-coding regions are captured and sequenced, followed by alignment to a reference genome, variant calling, and trio-aware analysis to assess inheritance patterns. This approach improves diagnostic yield by enabling precise filtering of candidate variants based on Mendelian segregation, making it especially effective for rare and undiagnosed conditions. Trio Exome suitable for investigation of rare variant association, adaptive TDT models and inferring transmission of inherited alleles.

Pros

- Higher diagnostic yield than singleton exome
- Accurate detection of individual variants
- Efficient filtering of benign variants
- Ideal for rare and undiagnosed diseases
- Strong support for clinical interpretation

Notes

- Requires samples from both biological parents + individuals
- Limited to coding and captured regions
- Structural variant detection is limited to indels
- Clinical validation recommended for key findings

Input



Genomic DNA (Germline)

DNA Quantity: 100 ng-1 µg

Sample Types



Blood



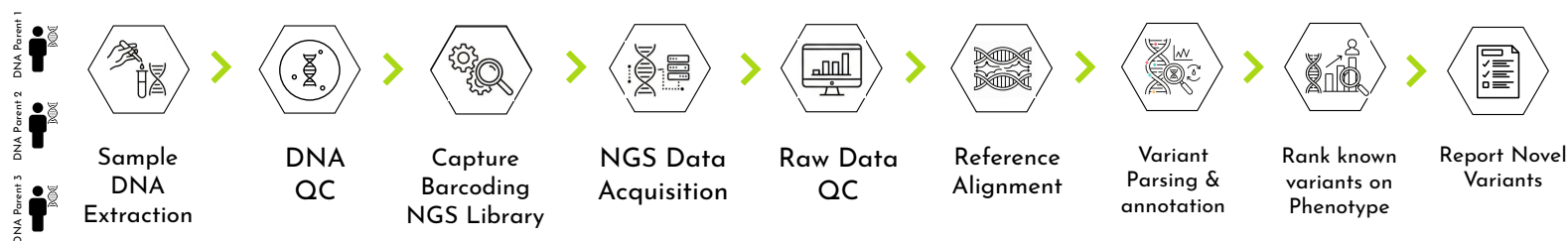
Tissue



Buccal
Swabs

Platform Illumina platforms | 150 * PE | 100 X Exome |
~35 MB x 3 x 24 GB

Process Map



Standard Deliverables

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary- trio
- Variant Summary Trio > Parent 1; Parent 2 > Individual
- Variant annotation and Variant effect prediction
- Novel Variants / Known Variants / Pathogenic Variants (if identified)
- Inheritance pattern analysis - Recessive/ Dominant disorders
- Phenotype based variant ranking (if scoped)

[Download Sample Report](#)

Applications

Method utilizes comprehensive genome-wide sequencing and analysis to enable hypothesis-driven and discovery-oriented research in human genomics. Long read based whole genome sequencing offers data suitable for detailed profiling of an individual genome. The state of the art method enable4s spanning repeat rich, High GC rich and complex region, without PCR biases to obtain data on genome organization, Structural variations, and genetic variations. The discovery genome pipeline enables germline haplotype phasing, methylation analysis and ranking of variants based on phenotypes providing deeper insights for genetic analysis.

Pros

- Enables unbiased genome-wide discovery
- Suitable for research discovery
- Detects rare and novel genetic/structural variants
- Scalable from small studies to large cohorts
- Supports integrative and population genomics
- Integrates epigenomic signatures and detailed variant annotations

Notes

- For research use only
- Interpretation depends on annotation of variants
- Denovo assembly not standard deliverable

Input



Genomic DNA (Germline)

- DNA Quantity: 200 ng-5 µg
- 5 µg High Molecular Weight DNA recommended

Sample Types



Blood



Tissue



Buccal
Swabs



Tumor
tissue

Platform ONT Long-read sequencing platforms

Process Map



Sample
DNA
Extraction



High Molecular
Weight DNA
QC



Library
Prep/QC



NGS Data
Acquisition



Data QC



Reference
Alignment



Variant
annotation



Reporting

Standard Deliverables

[Download Sample Report](#)

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Summary Trio ,SVP/Indel Zygosity, Coding, non-coding variant annotation for both Known and Novel
- Haplotype phasing with long reads
- DNA methylation analysis
- Pathogenic variant summary (if identified)
- CNV analysis and Variant effect prediction
- Structural variants / Repeat > Unaligned read
- *Note: Long read based local analysis or de-novo assembly to be scoped ; Not a standard deliverable



Targeted pathway interrogation (RUO)

Targeted pathway interrogation (RUO)

Applications

Targeted Pathway Interrogation (RUO) using NanoString nCounter[®] technology enables highly sensitive, amplification-free digital quantification of gene and miRNA expression. These curated panels provide robust, reproducible insights into key biological pathways, immune states, cancer biology, and host responses across diverse sample types, even from limited or degraded RNA (e.g., FFPE).. The use of specific probes allow consistent profiling of low-abundance transcript/ genes, that would need deep sequencing

Input



Total RNA

Conc: 100ng - 1 µg

DV 200 > 40%

Sample Types



FFPE Tissue
sections



Eukaryotic
cells



Tissue



Blood &
Body Fluids



Cancer
tissue

Platform Bruker nCounter Sprint Profiler

Available NanoString Panels

Panel Name	Recommended Samples /Batch	Targets /Total genes	Organism	Panel Description
PanCancer Progression Panel	12	800	Human	Profiles genes driving tumor growth, invasion, metastasis, angiogenesis, and extracellular matrix remodeling across cancer types.
PanCancer Pathways Panel	12	810	Human	Covers 13 core cancer pathways including cell cycle, apoptosis, PI3K, MAPK, DNA damage, and transcriptional regulation.
PanCancer Immune Profiling Panel	12	810	Human	Comprehensive immune profiling covering adaptive, innate, checkpoint, cytokine, and inflammatory signaling in the tumor microenvironment.
CAR-T Panel	12	780	Human	Evaluates CAR-T cell function, persistence, exhaustion, activation, and tumor-immune interactions in cell therapy studies.
Neuroinflammation Panel	12	740	Human / Mouse	Profiles immune and inflammatory pathways involved in neurodegeneration, neuroimmune crosstalk, and CNS disorders.
Human miRNA Panel	12	827 human miRNAs and 577 mouse miRNAs	Human	Quantifies human microRNAs involved in cancer, development, immune regulation, and disease-associated post-transcriptional control.
PanCancer IO 360 [™] Panel	12	770	Human	End-to-end immuno-oncology profiling including tumor biology, immune infiltration, checkpoint pathways, and therapeutic response markers.
Breast Cancer 360 [™] Panel	12	799	Human	Breast cancer-specific panel integrating tumor signaling, intrinsic subtypes, immune response, and predictive biomarkers.

Tumor Signaling 360™ Panel	12	800	Human	Focuses on oncogenic signaling networks driving tumor growth, survival, metabolism, and therapeutic resistance.
PAM50 Signature (RUO)	12	50	Human	Gold-standard 50-gene expression signature for intrinsic breast cancer subtyping and risk stratification.
ADC Panel	12	790	Human	Designed for antibody-drug conjugate studies, profiling target expression, payload sensitivity, and resistance mechanisms.
Metabolic Pathways Panel	12	768 (Human), 768 (Mouse)	Human / Mouse	Covers core metabolic pathways including glycolysis, TCA cycle, lipid metabolism, amino acid metabolism, and mitochondrial function.
miRNA Human	12	827 human miRNAs and 577 mouse miRNAs	Human / Mouse	High-throughput digital profiling of disease-associated and regulatory human microRNAs.
Custom Respiratory Virus Panel	12	Custom	Human / Viral	Customizable panel to profile host response and viral gene expression in respiratory viral infections.
Autoimmune Profiling Panel	12	770 (Human), 770 (Mouse)	Human/ Mouse	Profiles immune dysregulation, cytokine signaling, and inflammatory pathways implicated in autoimmune diseases.
Host Response Panel	12	785 (Human), 785 (Mouse)	Human / Mouse	Captures innate and adaptive immune responses to infection, inflammation, and tissue injury.
Human Organ Transplant Panel	12	770	Human	Monitors immune activation, rejection, tolerance, and inflammation in solid organ transplantation.
Immune Exhaustion Panel	12	797	Human / Mouse	Focuses on T-cell exhaustion, checkpoint signaling, dysfunction markers, and chronic immune activation.
Myeloid Innate Immunity Panel	12	770 (Human), 754 (Mouse)	Human / Mouse	Profiles macrophage, dendritic cell, neutrophil biology, and innate immune signaling pathways.

Standard Deliverables

[Download Sample Report](#)

- The project includes RNA QC,
- Probing for panel codeset. Hybridisation,
- Analyzing on the nanoString platform and providing the Gene Expression data/counts in .CSV file.
- Perform analysis on nSolver software to provide differentially expressed genes/transcript based on counts and normalized data with housekeeping reference genes.
- Graphical representation through heat map, histograms etc.

References

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- Richard V et al. Int J Mol Sci. 2023; PMID:36834918.
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- Ravichandran S et al. EBioMedicine. 2021; PMID:33906069.
- Tiwari S et al. Noncoding RNA Res. 2024; PMID:39131689.
- Shankar M et al. Sci Rep. 2023; doi:10.1038/s41598-023-47751-z.



Applied Genomics

Discovery of Variants & Polymorphisms

AP-DVPO-SRS

Applications

Genetic variation forms the foundation of phenotypic diversity, trait inheritance, and adaptive evolution. Theracues' Variant and Polymorphism Discovery service enables comprehensive identification of single nucleotide polymorphisms (SNPs), simple sequence repeats (SSRs), and repeat elements using high-throughput sequencing technologies, supporting applications in genomics, breeding, population genetics, and molecular marker development. The application support discovery of markers. Additional offerings offer primer + marker validation services.

Pros

- Genome-wide detection of SNPs, SSRs, and repeats
- Supports population and comparative studies
- Enables marker-assisted selection / discovery
- Applicable across multiple species

Notes

- Requires adequate sequencing depth per technology
- Reference genome enhances discovery of SNP's
- Polyploid genomes may need higher coverage
- Validation recommended for functional use

Input

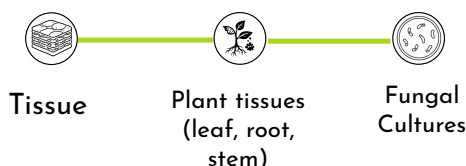


Genomic DNA

DNA Quantity: ≥200 ng - 1 μg (recommended)

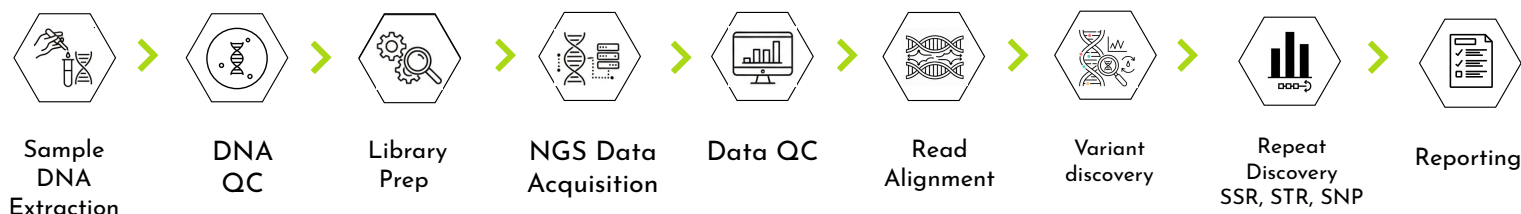
- High molecular weight DNA
- DNA QC-DIN

Sample Types



Platform Illumina / ONT Long-read sequencing platforms

Process Map



Standard Deliverables

- Sample QC and Summary of Methods
- Variation discovery, SSR, SNP, Indels
- Flanking primer and Variation sites report
- Comparative genomics for enriching SNP/SSR by group
- Optional primer set + Validation services

[Download Sample Report](#)

Rapid Molecular Diversity Analysis

AP-RAM0-DIA

Applications

Rapid Molecular Diversity Analysis enables quick and reliable assessment of genetic variation within and between populations. high-throughput sequencing. Supports diversity studies, germplasm characterization, breeding programs, and evolutionary research.

This service focuses on evaluating genetic diversity, population structure, and relatedness using molecular marker data. It enables rapid comparison of multiple samples to identify genetic similarities and differences, supporting applications in breeding selection, conservation genetics, and population-level studies. The method is suitable for characterization of new germplasm/non-model organisms with large genomes. The process leverages profiling of genetic markers, discovered and profiled using high throughput sequencing.

Pros

- Rapid turnaround time
- Efficient population-level analysis
- High reproducibility
- Supports breeding and conservation studies
- Scalable across sample sizes
- Not dependent on prior genome
- Suitable for novel genomes

Notes

- Marker discovered/profiled limit resolution
- Limited functional inference/coding genes profiled
- Suitable for studies involving population Structure/Breeding groups; Need K > 24 Individuals

Input



Genomic DNA



DNA Qubit > 1µg

- DNA Quantity: ≥ 100 ng (recommended)
- Gel QC/DIN - Tapestation 260/230 | 260/280

Sample Types



Blood/
PBMC



Tissue



Plant tissues
(leaf, root,
stem)



Feather/Hair
follicle/Gut DNA
droppings

Platform Illumina sequencing platforms

Process Map



Sample
DNA
Extraction



DNA
QC



Diversity
Ampli plex



Library
Prep



NGS Data
Acquisition



Data QC



Allele
Matrix



Population
Structure analysis



Diversity and
population
analysis

Standard Deliverables

- Sample QC and Summary of Methods
- Allele call matrix
- Population structure analysis
- Individual, Grouping/ Clustering

[Download Sample Report](#)

Cell line characterization - Clonality analysis

AP-CLCH-CLA

Applications

Cell line authentication is essential to ensure research validity, reproducibility, and data integrity. Genetic characterization of cell lines using sequencing, enabling reliable identification, authentication, and detection of cross-contamination. The method involves use of proprietary process to analyze the insertional; sites of transgene construct in complex genomes such as CHO. The process enables identification of insertion copies. Identification of transgene insert integrity and ensure Master Cell Bank/ Working Cell Bank clones encode mutations free proteins. The process also uses whole genome SNP and CNV analysis to match the host genome and ensure it is free of any viral integration events.

The report can provide data for filing IP on clones, provide characterization for regulatory filing.

Pros

- High accuracy and reproducibility
- Rapid and cost-effective
- Detects cross-contamination
- Suitable for routine QC
- Low DNA input required

Notes

- Limited to targeted loci
- Reference profiles required
- Not genome-wide
- Periodic testing recommended

Input



Genomic DNA

- DNA Quantity: ≥ 100 ng - 1 μ g (recommended)
- High Molecular Weight DNA
- Requires information of vectors used for transgenesis

Sample Types



Human
Cell-lines



Animal
Cell-lines



Fresh or
frozen cell
pellets

Platform ONT Long-read Sequencing Platforms

Process Map



Standard Deliverables

[Download Sample Report](#)

- Reporting Summary of method / Reference Genomes
- Site of integration of Transgenes
- Host genome CNV / SNP based similarity map
- Copy number of transgene- Count of sites of integration
- Abbrent Integration- Mutation/ Indels/ Truncation/ Deletion

Hybridoma sequencing

AP-HYBM-SEQ

Applications

The hybridoma sequencing application aims to recover heavy and light chains from mouse, rat, and rabbit hybridoma. Hybridomas can encode multiple antibody transcripts, and in many instances express aberrant chain or multiple forms.

We offer chain annotation, Flagging of aberrant chains and provide add-on services to predict functional pairs using in-house data models, including gene-synthesis to re-construct the antibody V_H and V_L chains for expression. This service focuses on sequencing antibody variable domains (VH and VL) hybridoma samples. By combining targeted amplification with next-generation sequencing, it enables reliable recovery of functional antibody sequences for cloning, expression, humanization, and intellectual property protection.

Hybridoma sequencing ensures long-term preservation of antibody sequence information independent of cell line stability.

Pros

- Accurate VH/VL sequence recovery
- Supports antibody cloning / Expression workflow
- Preserves antibody IP / Engineering

Notes

- Requires high-quality RNA/DNA
- Mixed clones may complicate analysis
- Recovered clone will need functional authentication

Input



Total RNA

- RNA: 200 ng - 500 ng

Sample Types



Hybridoma
cell pellets



Cultured
hybridoma
cells

Platform ONT Long-read sequencing platforms

Process Map



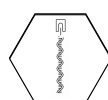
Sample
Prep



RNA
QC



cDNA Target
Amplification



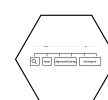
Library
Prep



NGS Data
Acquisition



Data QC



Clonotype
Identification
Pipeline



VH VL
Sequencing

Standard Deliverables

- Sample QC Report, Summary of methods
- NGS Library summary
- VH / VL- Summary and sequence of chains
- Annotation of clonotypes/ flagging aberrant / non-productive chains

Add-ons

- In-Silico VH-VL pairing
- Gene reconstruction and synthesis
- Flagging aberrant chain
- In-Silico pairing
- Gene synthesis services

[Download Sample Report](#)

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